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Deputy Directorate General of Quality Control

Plasma Smelting Laboratory-3-H

TEST REPORT

Test Report No: TR (024)/PS/22

Job Card No: JC (044)/PS/22

Date: 27-04-22

INDENTER'S DATA

Name: Fraz-ur-Rehman	Designation: Manager
Group & Site: MP/3C	Directorate: ADP
Organization: NDC	Contact No: 3627

GENERAL DATA

Sample Name: Silica Phenolic	ID No: PP-10A-AbI-02 (Sample 02)		
Lot No: Si (820)-10A/1120-221~333	Lot Size: 8000 RM		
Quantity: 01	Chemical Composition: Composite		
Test Type: General	Sample Receiving Date: 26-04-22	Test Date: 26-04-22	

TEST DATA

Test Name: Ablation Test	
Qualification Standard: NDC-AbI/QC/Si-C Ablatives/02 (Rev#1)	
Standard Test Method: 1-E 458-72(Re approved 2002)	
Test Environmental Conditions: Ambient	
Test/Calorimetry Equipment & Instruments Used: Huels Arc Heater, Digital Micrometer.	
Radiation & Optical Equipment used: Nil	Enthalpy Measurement Technique: By Empirical Formula
Flow Temperature:	3000 K
Flow Enthalpy:	3.526 MJ/Kg
Duration of Exposure:	20 Sec
Results:	▪ Test results attached (at page 02 of 02)
Remarks:	▪ These results relates only to the sample mentioned and tested above. ▪ This test report shall not be reproduced except in full, without written approval of the laboratory. ▪ The working medium for generation of Plasma Arc is air. ▪ Uncertainty in ablation rate calculation is ± 0.00774 mm/Sec at 95 % confidence level.

Test Performed By: Wasim Ahmed Zahid, GM (Elect) *Wasim Zahid 27.4.22*Reviewed By: Muhammad Zahid Irfan, PD (PS) *Muhammad Zahid Irfan 27.4.22*

Contact No: 3205/3287

Approved By: Khalid Atta, Dir (STE) *Khalid Atta 28.4.22*

Prepared By: FM/GM

Reviewed & Recommended By: GM/HOD

Approved By: GMR/HOD/PD

PS/FM/02, Issue 20(09-07-21)

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Plasma Smelting Laboratory-3-H

TEST REPORT

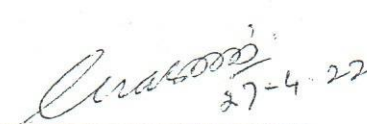
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Test Results

Sr #	Description	Measured Value	Qualification Criteria Standard Value	Status
1	Average initial thickness before test	10.948 mm	NA	Qualified
2	Final residual thickness after test	10.908 mm	NA	
3	Maximum ablation in terms of thickness	0.04 mm	NA	
4	Ablation Rate in terms of thickness	0.002 mm/Sec	≤ 0.15 mm / Sec	
5	Max Back Surface Temperature	NA	NA	

Additional Remarks: Nil

Test Performed By: 

Wasim Ahmed Zahid, GM (Elect)

End of Report

Prepared By: FM/GM	Reviewed & Recommended By: GM/HOD	Approved By: GMR/HOD/PD
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PS/FM/02, Issue 20(09-07-21)